

BUMJUN PARK

Ph.D. Student in Biostatistics

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EDUCATION

Sep 2023–(present) (expected Aug 2028)	Ph.D., Biostatistics, <i>University of Washington</i>
Sep 2018–May 2023	B.S., Statistics, <i>University of Wisconsin-Madison</i> (certificate in Mathematics & Economic Analytics)

PUBLICATIONS

(* denotes corresponding author)

2025	Lin, X.*, Park, B. , Zahasky, C. & Kang, H. (2025). "Semiparametric Spatial Point Processes". <i>arXiv</i> . (manuscript)
2024	Park, B.* , Kang, H., & Zahasky, C. (2024). "Statistical Mapping of PFOA and PFOS in Ground-water Throughout the Contiguous United States". <i>Environmental Science & Technology</i> , 58, 44, 19843–19850. (manuscript) (code)

RESEARCH EXPERIENCE

Sep 2025 – (present)	Research Assistant , <i>University of Washington</i>	Advisor: Amy Willis
	• Conducted theoretical and methodological research on nonparametric regression methods incorporating known graph structures such as phylogenetic trees • Maintained and extended R packages, including issue resolution and feature development on GitHub.	
Dec 2024 – (present)	Independent Study , <i>University of Washington</i>	Advisor: Jon Wakefield
	• Conducted a critical evaluation of the Log-Quad model for estimating under-five mortality rates, assessing its theoretical assumptions and empirical performance through simulation studies. • Investigated a Bayesian method of using granular data to construct informative priors for estimating hazard functions for countries with coarser data.	

Sep 2023 – (present)	Research Assistant , <i>University of Washington</i>	Advisor: Eardi Lila
	• Studied and investigated multivariate functional data analysis methods for predicting ischemic strokes, working with the research group under Professor Mahmud Mossa-Basha, Department of Radiology. • Developed and implemented a method of modeling Alzheimer's disease progression through the application of differential equations and rate-based analysis using longitudinal biomarker data.	
Sep 2024 – Sep 2025	Research Assistant , <i>Fred Hutch Cancer Center</i>	Advisor: Jing Ma
	• Studied the impact of compositionality and false discovery rate (FDR) control on microbiome network estimation through simulation and methodological evaluation. • Developed and reviewed statistical models for compositional and graphical analyses of microbiome data	
Sep 2022 – May 2023	Data Analyst , <i>Nelson Institute, UW-Madison</i>	Advisor: Jonathan Patz
	• Preprocessed and analyzed data for environmental policy, air quality, and epidemiology projects. • Applied spatial random forest models to investigate the relationship between malaria prevalence and factors such as vegetation coverage, insecticide-treated net distribution, precipitation, and livestock populations in Kenya.	
May 2022 – May 2023	Undergraduate Research Assistant , <i>UW-Madison</i>	Advisor: Chris Zahasky
	• Implemented web-scraping algorithms to collect PFAS concentration data from sources such as the U.S. Air Force and Wisconsin Department of Natural Resources. • Developed geostatistical visualizations and built an Inhomogeneous Poisson Process model to predict PFAS concentrations while accounting for opportunistic sampling.	
Feb 2022 – May 2023	Undergraduate Research Assistant , <i>UW-Madison</i>	Advisor: Stephen Gammie
	• Analyzed RNA-sequencing gene expression data from Alzheimer's disease models to identify differentially expressed genes. • Processed gene expression data for Alzheimer's and Parkinson's disease patients, implementing machine learning models to classify diseases using top-scoring differential gene pairs.	

TEACHING EXPERIENCE

Jan 2026 – Mar 2026	BIOST515: Applied Biostatistics II , <i>University of Washington</i> <i>Lead Instructor: Amy Willis</i>	(Teaching Assitant)
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PRESENTATIONS

May 2025	UW Biostat Student-Invited Speaker Poster Session (<i>Prof. Emmanuel Candès</i>) (Poster) Park, B. , Lila, E., Mossa-Basha, M. "Tracking Vessel Wall Changes over Time"
Jan 2025	UW Biostat Student Seminar (Talk) Park, B. "False Discovery Rate and Multiple Testing: Detecting Microbiome Networks"
Oct 2024	UW Biostat Student Seminar (Talk) Park, B. "Function on Function Regression on Sparse Observations: Multivariate Functional PCA on Vessel Wall Imaging Data"
Mar 2023	AWRA Wisconsin Section (Poster) Park, B. , Kang, H., Gnesda, W., & Zahasky, C. "Groundwater Contamination of Per- and Polyfluoroalkyl Substances in the United States - Insights from an Ecological Sampling Bias Correction Method"